

Quantum Mechanics Homework 4

Lectures 8–9

Instructions. Show the main steps of your calculation. You may leave answers in terms of $\hbar$, m , and the parameters given in each problem. Each problem is worth 20 points.

Problem 1. Energy superposition and unitary evolution [20 points]

Let $\phi_1(x)$ and $\phi_2(x)$ be orthonormal energy eigenfunctions:

$$\hat{H}\phi_1 = E_1\phi_1, \quad \hat{H}\phi_2 = E_2\phi_2, \quad E_1 \neq E_2, \quad (1)$$

with

$$\int_{-\infty}^{\infty} \phi_m^*(x)\phi_n(x) dx = \delta_{mn}. \quad (2)$$

At $t = 0$, the state is

$$\psi(x, 0) = \frac{3}{5}\phi_1(x) + \frac{4i}{5}\phi_2(x). \quad (3)$$

- Verify directly, using the integral above, that $\psi(x, 0)$ is normalized.
- Write the state $\psi(x, t)$ at a later time.
- Find the probabilities of measuring E_1 and E_2 .
- Show that the total probability remains one for all time. Is this state stationary? Briefly explain.

Problem 2. Interference between two energies [20 points]

Suppose $\phi_1(x)$ and $\phi_2(x)$ are real orthonormal energy eigenfunctions, and

$$\psi(x, t) = \frac{1}{\sqrt{2}} [\phi_1(x)e^{-iE_1t/\hbar} + \phi_2(x)e^{-iE_2t/\hbar}], \quad E_1 \neq E_2. \quad (4)$$

- Calculate the position probability density $|\psi(x, t)|^2$. Keep the interference term.
- What relative phase develops between the two energy components?
- Find the period of the oscillation in the interference term.
- The position probability density can change with time. Explain why the probabilities of measuring E_1 and E_2 nevertheless remain constant.

Problem 3. A square well with constant potential energy [20 points]

Consider a particle confined to the infinite square well

$$V(x) = \begin{cases} V_0, & 0 < x < L, \\ \infty, & x \leq 0 \text{ or } x \geq L, \end{cases} \quad (5)$$

where V_0 is a constant. The energy eigenfunction satisfies the boundary conditions

$$\phi(0) = 0, \quad \phi(L) = 0. \quad (6)$$

Inside the well, you may use

$$\phi(x) = A \cos(kx) + B \sin(kx), \quad E = V_0 + \frac{\hbar^2 k^2}{2m}. \quad (7)$$

- (a) Use $\phi(0) = 0$ to determine the coefficient A .
- (b) Use $\phi(L) = 0$ to show that the allowed wavenumbers are

$$k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots \quad (8)$$

- (c) Find the allowed energies E_n . Write E_1 and E_2 explicitly.
- (d) The normalization factor is $\sqrt{2/L}$. Write the first two normalized eigenfunctions inside the well. Does the constant V_0 change the eigenfunctions, or does it only shift the energies?

Problem 4. Infinite square well with symmetric boundaries [20 points]

Consider a particle in the infinite square well

$$V(x) = \begin{cases} 0, & -a < x < a, \\ \infty, & |x| \geq a. \end{cases} \quad (9)$$

The boundary conditions are

$$\phi(-a) = \phi(a) = 0. \quad (10)$$

- (a) Inside the well, show that

$$\phi''(x) + k^2\phi(x) = 0, \quad k^2 = \frac{2mE}{\hbar^2}, \quad (11)$$

and write its general solution.

- (b) Apply the two boundary conditions and show that the allowed wavenumbers are

$$k_n = \frac{n\pi}{2a}, \quad n = 1, 2, 3, \dots \quad (12)$$

Find the corresponding energies E_n .

- (c) Write the first three normalized energy eigenfunctions inside the well. State that they are zero outside the well.
- (d) For each of the first three states, give its parity and number of interior nodes. Also find the ratio $E_1 : E_2 : E_3$.

Hint: You may shift the coordinate using $y = x + a$, so that the interval $-a < x < a$ becomes $0 < y < 2a$.